

# Quantum Benchmarks

## Hardware Runs

Report issue date: May 4, 2026

This benchmarking report was generated using Classiq v1.10.1.

The report includes hardware runs for a collection of pre-defined and user-defined functions and algorithms, where each benchmark can be instantiated at different problem sizes. For each benchmark experiment, we report a score defined specifically for that example, together with the execution time<sup>1</sup> and quantum-program properties obtained through Classiq's hardware-aware synthesis<sup>2</sup>. For further information, see the Benchmarking directory in classiq-library.

---

<sup>1</sup>Currently, the reported time includes both execution time and waiting time.

<sup>2</sup>The properties of the actual circuit executed on the hardware may differ slightly due to pre-run transpilation.

## Quantum Volume

We benchmark Quantum Volume following Cross et al., PRA 100, 032328 (2019). For each width  $n$ , we generate square model circuits of width  $n$  and depth  $n$ , and measure the heavy-output probability  $h$ , namely the total probability assigned to outcomes whose ideal probabilities are above the median of the ideal output distribution.

A width is considered to pass if the heavy-output probability exceeds the threshold  $2/3$  according to a statistical lower bound. For runs with at least 100 completed trials, we use the pooled confidence criterion of Cross et al. If  $n_c$  is the number of completed circuits,  $n_s$  the number of shots per circuit,  $n_h$  the total number of heavy outputs observed across all trials, and  $\sigma$  the chosen confidence parameter, we use

$$h_{\text{LB}} = \frac{n_h - \sigma \sqrt{n_h(n_c n_s - n_h/n_c)}}{n_c n_s},$$

and declare the width successful when  $h_{\text{LB}} > 2/3$ . For smaller runs, we use the simpler normal approximation

$$h_{\text{LB}} = \bar{h} - \sigma \text{SE}(\bar{h}),$$

where  $\bar{h}$  is the sample mean heavy-output probability across trials and  $\text{SE}(\bar{h})$  is its standard error.

The reported Quantum Volume is

$$QV = 2^{n_{\text{max}}},$$

where  $n_{\text{max}}$  is the largest tested width that passes. Consequently, if the largest executed width also passes, the true quantum volume is at least this value and may in fact be larger.

For more details see Quantum Volume notebook.

### 2–4 qubits (10 trials, 1000 shots)

| Provider | Backend               | Quantum Volume | Trials | Average Elapsed Time (min) |
|----------|-----------------------|----------------|--------|----------------------------|
| Classiq  | simulator             | 16             | 30/30  | 0.01                       |
| Classiq  | simulator_statevector | 16             | 30/30  | 0.01                       |